SURJECTIVITY OF THE POINT MAP OF A CARTESIAN DIAGRAM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \\ Z & \longrightarrow & Y \end{array}$$

be a cartesian diagram of algebraic spaces over S. Then the map of sets of points

$$|Z \times_Y X| \longrightarrow |Z| \times_{|Y|} |X|$$

is surjective.

Proof. Namely, suppose given fields K, L and morphisms $\mathrm{Spec}(K) \to X$, $\mathrm{Spec}(L) \to Z$, then the assumption that they agree as elements of $|Y|$ means that there is a common extension M/K and M/L such that $\mathrm{Spec}(M) \to \mathrm{Spec}(K) \to X \to Y$ and $\mathrm{Spec}(M) \to \mathrm{Spec}(L) \to Z \to Y$ agree. And this is exactly the condition which says you get a morphism $\mathrm{Spec}(M) \to Z \times_Y X$. □

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